Friday Flex — Chapter 4: Energy Transfer

> Friday = FLEX only. No essential new content. Nothing here is required for Monday.

WEEK 4 (Sound + Light)

RECOVER (finish + catch up)

- **Finish sound/light evidence** — complete any unfinished observation tables.
- **Notebook catch-up** — Insert 1 (energy transfer) and Insert 2 (sound + light).
- **Reader recovery** — reread Ch.4 pp. 17–19 (transfer, sound, light).

PRACTICE — Energy Path Sort (CUSTOM)

- **Time:** 10–15 min · **Materials:** printed cards (source/target pairs) · **Independence:** guided
- Sort cards into SOURCE → ENERGY MOVES → TARGET for: bell→paper · flashlight→card · sun→your skin · speaker→your ears.
- Extension: write the evidence (the change) on the back of each card.
- **Concepts reinforced:** source/target model, evidence of transfer.

PRACTICE — Transfer Detective (CUSTOM)

- **Time:** 10 min · **Materials:** worksheet or whiteboard · **Independence:** independent
- Given 5 mini-scenarios, students circle the source and underline the change (evidence).
- **Concepts reinforced:** identifying evidence of transfer.

EXTEND — Sound / Light Mystery Files (EXTENSION)

- **Time:** 15 min · **Materials:** printed cards · **Independence:** independent
- "Mystery" scenarios: a window rattles when a truck passes; a solar calculator works in sunlight; a shadow appears when you block a flashlight. Students name the source, the change, and whether sound or light transferred energy.
- **Concepts reinforced:** transfer inference from evidence.

EXTEND — Flashlight Challenge (EXTENSION)

- **Time:** 10 min · **Materials:** flashlight + barrier cards · **Independence:** guided
- How many barriers before the target stops lighting? Predict, test, explain.

WEEK 5 (Heat + Electricity + Synthesis + CER)

RECOVER (finish + catch up)

- **CER finish** — complete AP 9.2-style claim ("I know ___ because ___").
- **Notebook completion** — Inserts 3 (heat + electricity) and 4 (transfer vs transformation) and 5 (evidence + CER).
- **Missed-lab recovery** — rerun the heat or electricity setup for absent students (teacher-led).

PRACTICE — Transfer / Transformation / BOTH Sort (CUSTOM)

- **Time:** 15 min · **Materials:** printed cards · **Independence:** guided
- Sort: flashlight, buzzer, motor, lamp, warm-water setup, sound system → TRANSFER / TRANSFORMATION / BOTH.
- Rule: BOTH is allowed. No false either/or.
- **Concepts reinforced:** transfer vs transformation vs BOTH.

PRACTICE — Energy Chain Error Detective (CUSTOM)

- **Time:** 10 min · **Materials:** worksheet · **Independence:** independent
- For each chain (e.g., battery → flashlight → light), students: (1) find where energy transferred, (2) find where transformation occurred, (3) fix a planted false statement (e.g., "energy disappeared").
- **Concepts reinforced:** transfer + transformation can happen together; energy isn't destroyed.

EXTEND — Everyday Device Cards (EXTENSION)

- **Time:** 15 min · **Materials:** printed cards · **Independence:** guided
- Cards: flashlight, fan, speaker, wind-up toy, toaster, solar calculator. For each: what moves? what changes form? transfer / transformation / BOTH? (wording verified before use).
- **Concepts reinforced:** BOTH classification in everyday systems.

EXTEND — Low-Voltage Circuit Challenge (EXTENSION)

- **Time:** 15 min · **Materials:** battery holder + wires + bulb/buzzer/motor (low-voltage only) · **Independence:** guided
- Make the bulb light AND the buzzer sound together; name the evidence of transfer in each.
- **Concepts reinforced:** electric-current transfer, evidence of change.

Rule check (both weeks)

- [x] Not required for Monday's new core instruction
- [x] Low-prep, quick reset (end-of-day prep)
- [x] Low-voltage electricity only · no hot plates · no wall outlets